

Module 26 - Hoisting (Variables & Functions)

1. What is Hoisting?

Hoisting is a JavaScript mechanism where variable and function declarations are moved to the top of their scope during the compilation phase.

2. Variable Hoisting

- **var** → Declaration is hoisted and initialized with **undefined**.
- **let / const** → Declaration is hoisted but stays in the **Temporal Dead Zone (TDZ)** until initialization.

3. Function Hoisting

- Function declarations are fully hoisted.
- They can be called before their declaration in the code.

4. Function Expression Hoisting

- A function stored in a variable is not fully hoisted.
- With **var** → the variable is undefined before assignment, so calling it causes a **TypeError**.
- With **let/const** → accessing it before initialization causes a **ReferenceError**.

Example Code:

```
// Variable Declaration Hoisting

/* Before Code Execution - Declaration will come at top

var a    → declaration is hoisted

let b    → declaration is hoisted but stays in TDZ

const c  → declaration is hoisted but stays in TDZ

sample() → function declaration is fully hoisted

sample1  → declaration depends on var/let

*/

// var is hoisted with the value undefined
console.log(a);
var a = 10;
console.log(a);

// let is in the Temporal Dead Zone (TDZ)
console.log(b); // ReferenceError
let b = 20;
console.log(b);
```

```
// const is in the Temporal Dead Zone (TDZ)
console.log(c); // ReferenceError
const c = 30;
console.log(c);

// Function Declaration Hoisting
// Function declaration can be called before its definition.
sample();
function sample(){
    console.log("IM in");
}

// Function Expression Hoisting
// let sample1
// var sample1
console.log(sample1);
sample1();

// var function expression gives TypeError when called before assignment.
// var sample1 = function(){
//     console.log("Am i accessible");
// }

// let function expression gives ReferenceError before initialization.
let sample1 = function(){
    console.log("Am i accessible");
}
sample1();
```

Hoisting Comparison Table

Feature	var	let	const	Function Declaration	Function Expression
 Hoisted ?	✓ Yes	✓ Yes	✓ Yes	✓ Yes (Fully)	✗ No
 Initialized ?	✓ Yes (as undefined)	✗ No	✗ No	✗ No (until definition)	✗ No
 Accessible Before Declaration ?	✓ Yes	✗ No	✗ No	✓ Yes	✗ No
 Error Before Declaration ?	✗ No	✓ Yes (ReferenceError)	✓ Yes (ReferenceError)	✗ No	✓ Yes (TypeError when called)
 TDZ (Temporal Dead Zone)	✗ No	✓ Yes	✓ Yes	✗ No	N/A